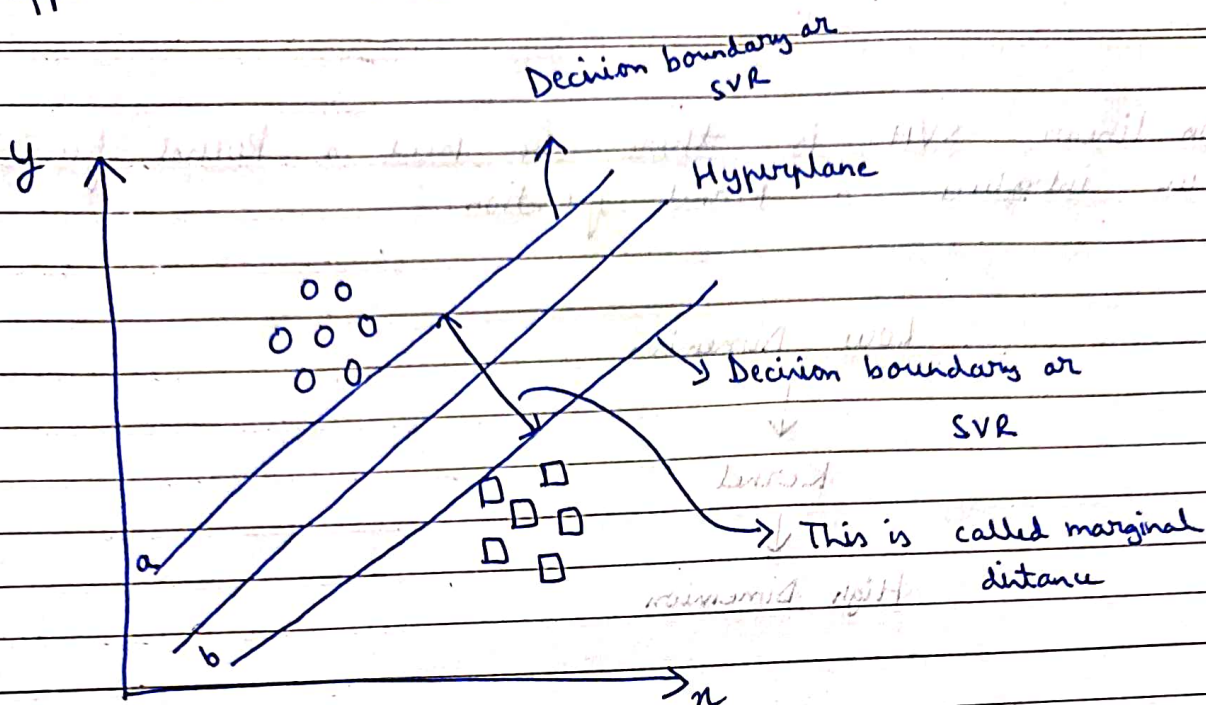


* Support Vector Machine

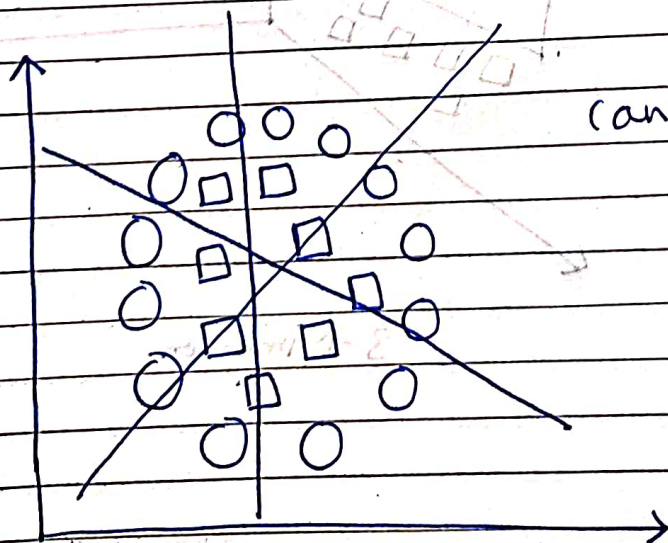
①



Linear SVM

The distance between a and b should be maximum

②



cannot divide the two classes doesn't matter which line you plot

This is called Non Linear SVM

If non linear SVM is there we need a kernel function
or we introduce a kernel function.

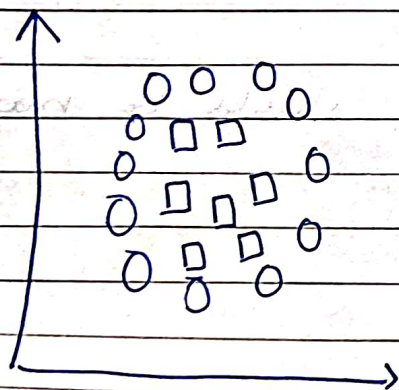
Low Dimension



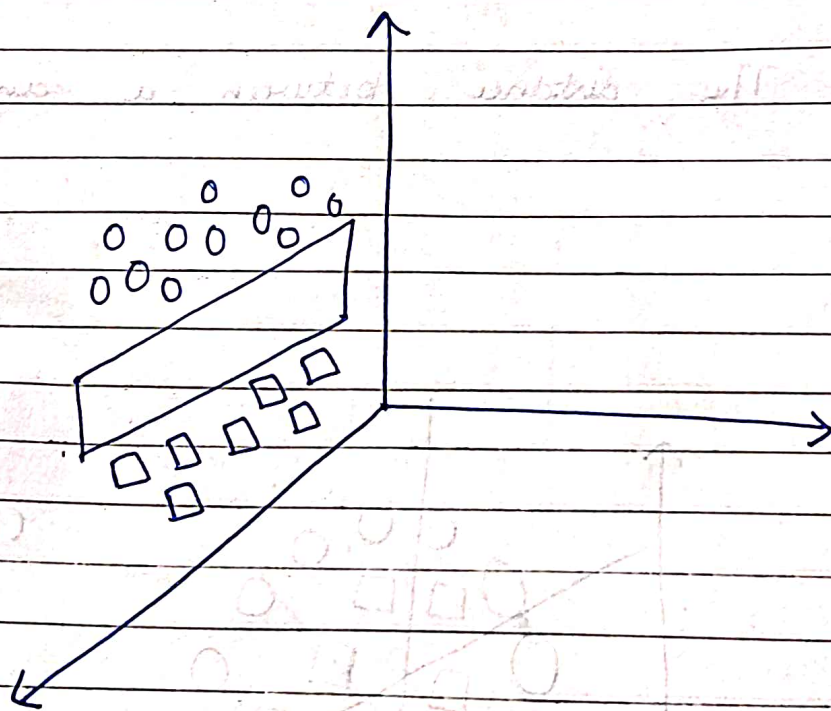
Kernel



High Dimension



2-Dimension



3-Dimension.